

M1chrome — Stellar-Chrome Dielectric Coating V2.0

Expanded Aegis Foundational Research: Metallic-Look Dielectric Coating System

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- 6-layer Stellar-Chrome stack fully specified (MAX Phase primer to QD YInMn topcoat)
- Optical performance: 37% heat load reduction vs chrome under X-class flare conditions
- Flash sintering ZrO_2 at 1150°C : $\approx 150^\circ\text{C}$ vs conventional (2025 data)
- MXene FSS tile integration: 92 dB SE while maintaining metallic visual appearance

1. The Stellar-Chrome Concept

YInMn Blue is the primary color of the Aegis system — but many architectural and automotive applications require a metallic chrome aesthetic. Standard chrome uses electrodeposited metallic chromium — a highly conductive GIC susceptor.

Stellar-Chrome achieves the optical appearance of chrome while maintaining complete dielectric isolation:

6-Layer Stellar-Chrome Stack: 1. MAX Phase Ti_3AlC_2 primer (0.5 mm) — machinable base, mechanical adhesion, electrically isolated via PEEK undercoat 2. ZrO_2 base coat (50 μm , flash sintered at 1150°C) — white reflective base 3. MXene $\text{Ti}_3\text{C}_2\text{T}_x$ FSS tiles (50 μm , 3×3 cm discontinuous) — 92 dB EMI, metallic sheen 4. SiO_2 optical spacer (200 nm) — constructive interference for silver-metallic color 5. ZrO_2 cap coat (30 nm) — durability, prevents MXene oxidation 6. QD-YInMn Blue topcoat (500 nm) — UV/NIR rejection superimposed on chrome appearance

Visual result: Silver-chrome appearance in visible light + 85% NIR rejection + 92 dB EMI SE.

2. Thermal Performance Under X-Class Solar Flare

For an X-class flare delivering 1 MW/m^2 NIR flux: - Standard chrome: absorbs 95% = 950 kW/m^2 $\rightarrow$ surface vaporization - Stellar-Chrome: reflects 85% = 850 kW/m^2 rejected $\rightarrow$ 150 kW/m^2 absorbed $\rightarrow$ 37% heat load reduction vs chrome (37% of chrome's damage)